Problem 7

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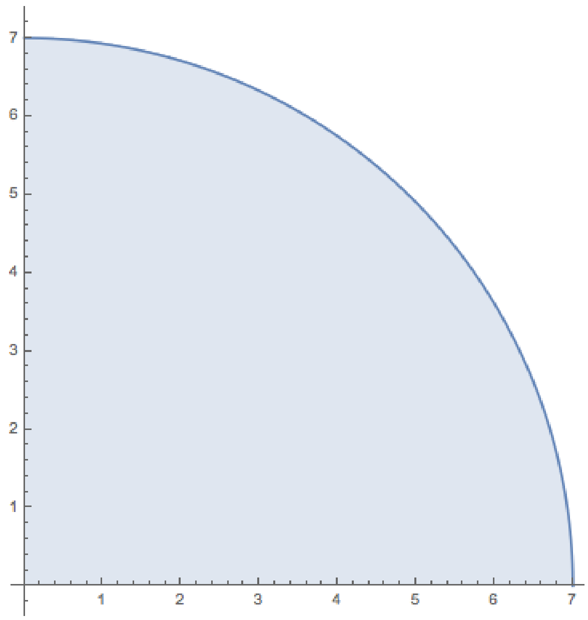
Problem 1 *A part of a circle is shown in the figure.*



(a) *If we express the area of the shaded region in the figure as the limit of Riemann sums*

$$A = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x,$$

find the function f *and the interval* $[a, b]$.

Explanation. $f(x) = \sqrt{49 - x^2}$, and the interval $= [0, 7]$.

(b) *Compute the limit.*

$$\lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{49 - (x_k^*)^2} \cdot \frac{7}{n}$$

Explanation. $\lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{49 - (x_k^*)^2} \cdot \frac{7}{n} = \frac{49\pi}{4}$